

TSO – Time-Sharing Option

Introduction





WHAT IS TIME SHARING?

Time-sharing enables more than one user of a computer system to access the system concurrently without interfering with each other. In the early days of mainframes, the systems could only execute one task at a time. Since human interactions with the system take far more time than internal computing operations, when the system allows users to take turns, each user may feel as if they have the whole system to themselves – at least, most of the time.

Computers spend far more time waiting for us to decide what to do next than we spend waiting for them to complete whatever internal processing they are performing for us.



THE TIME-SHARING OPTION



When the IBM System/360 was introduced (1964), the standard input device was a punched card reader like the museum piece shown at left. The standard input stream was, and still is, the SYSIN data set. To this day, SYSIN contains 80-character text records that duplicate the layout of 80-column punched cards.

Although these were batch processing systems, it wasn't long before customers demanded interactive support. In 1971, when IBM released the Time-Sharing Option (TSO), they still believed most customers would be satisfied running punched cards in batch mode. It was offered as an add-on product.

TSO didn't become a part of the base operating system configuration until the advent of z/OS in 2001. The product name still includes the word, "option."

BLOCK-MODE TERMINALS AND AID KEYS

We mentioned 3270 terminal emulation when we were setting up the environment for the bootcamp. We'll talk more about this in the module about CICS. For now, just know that this type of device was state-of-the-art at the time TSO came out. It became the standard terminal for the IBM mainframe line.

This is a block-mode terminal, as opposed to character-mode devices such as the DEC VT-100. In a nutshell, that means that only certain keys caused data to be transmitted to the host. The other keys updated the contents of a buffer located in the terminal. The keys that sent data to the host are called *attention identifier* keys, or AID keys.

The AID keys are Enter, Clear, System Request, Program Attention (PA) 1 and 2, and Program Function 1 through 12 (up to 24 on later models). Several of these don't exist on standard keyboards today. Your terminal emulator software will offer a way to simulate these keys when you're using TSO.



TERMINAL EMULATOR MENU

```
File  Options  Keypad

IBM z/OS Standard Development/Test Image
Built on: 2024-03-21-16:35
System:      VS01
Workstation IP: 71.226.56.53
Date:        01/26/26
Time:        10:27:34

To logon to an Application type one of:
--> LOGON XXX (XXX is Vtam APPL)
--> TSO userid
--> CICSTS61
--> IMS15APL
--> CNM01

ENTER CHOICE ==> █
```

The special keys on 3270 terminals are not present on modern keyboards.

The brain-in-a-jar thinks it's talking to a real 3270 terminal, so we need a way to transmit the special key codes to it.

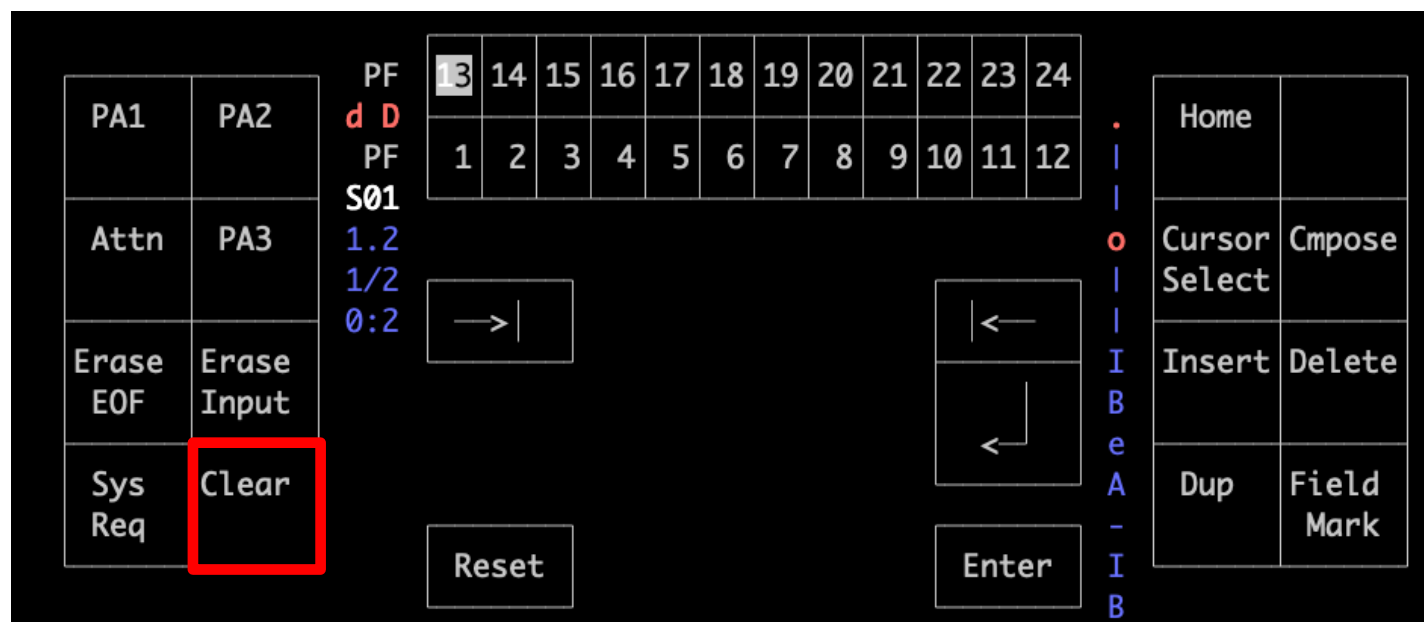
Terminal emulation software provides a way to simulate these keys. Each product has its own implementation of this.

The product I use is c3270, a command-line wrapper for the open source emulator, x3270. It provides a menu bar at the top of the screen, as shown at left.

When I need one of the special 3270 keys, I choose the Keypad option on that menu.

```
/*
(-----) ||-|| |□□□□|
||||||| ||-|| |□□□□|
```

TERMINAL EMULATOR KEYPAD

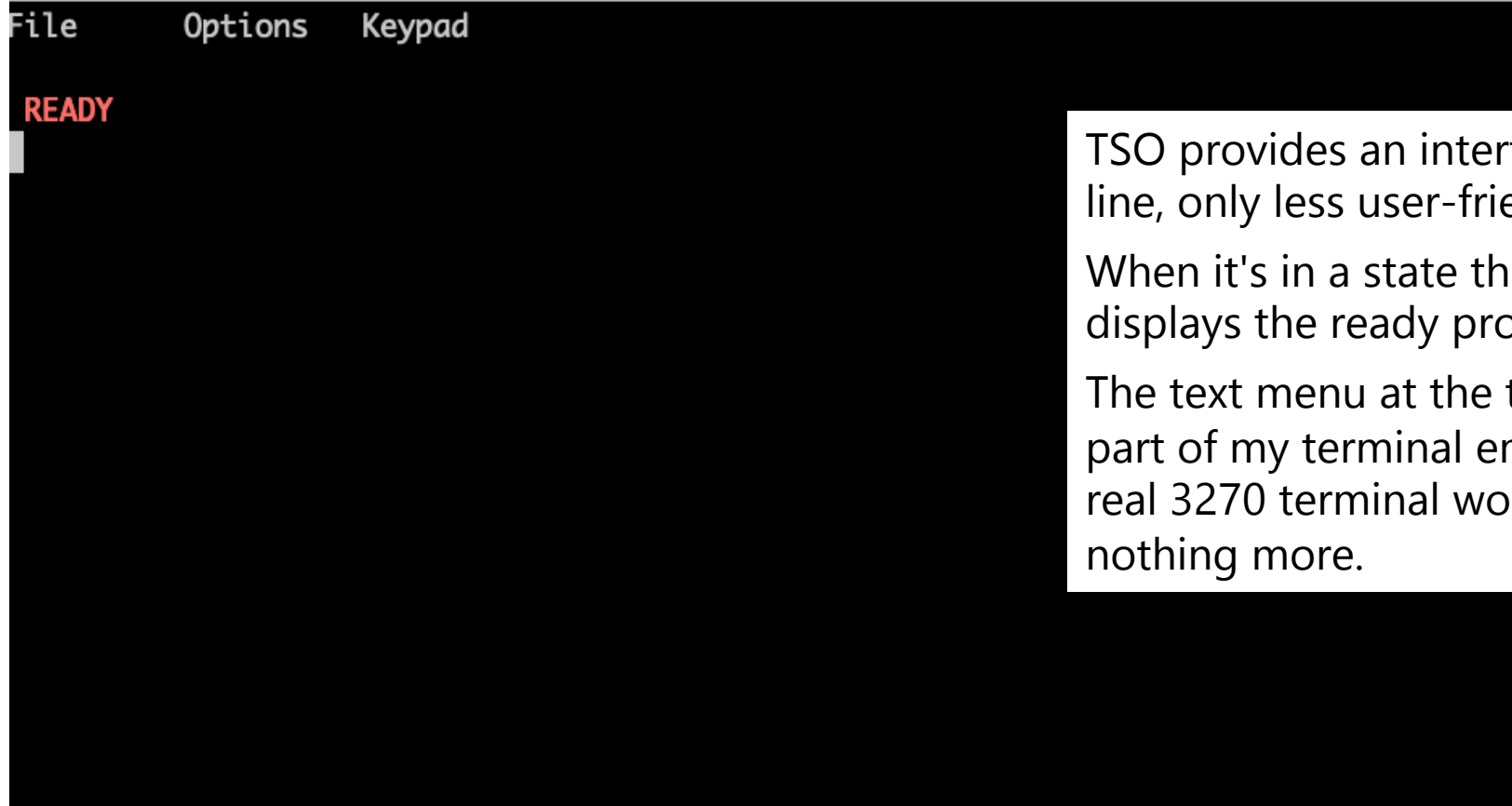


The Keypad option overlays a virtual keypad on the screen. A key can be selected with a mouse click. The emulator transmits the corresponding key code to the mainframe.

For instance, if I need to simulate pressing the Clear key on a 3270 keyboard, I click on the box labeled "Clear" on the virtual keypad.

This is only an example. The terminal emulator you're using may implement this functionality differently, but the special keys are available somewhere in the product's user interface.

THE TSO READY PROMPT



TSO provides an interface similar to a command line, only less user-friendly.

When it's in a state that can accept input, TSO displays the ready prompt.

The text menu at the top of the screen shot is part of my terminal emulator display. TSO on a real 3270 terminal would just say "ready" and nothing more.

THE TSO READY PROMPT

```
File      Options  Keypad

READY
time
IKJ56650I TIME-05:09:12 PM. CPU-00:00:00 SERVICE-1814 SESSION-00:03:23 FEBRUARY
16,2024
READY
█
```

We can enter a TSO command now. The command, TIME, will display the system time. When the command finishes, TSO displays the ready prompt again.

THE TSO READY PROMPT

```
File      Options  Keypad

READY
time
IKJ56650I TIME-05:09:12 PM. CPU-00:00:00 SERVICE-1814 SESSION-00:03:23 FEBRUARY
16,2024
READY
profile
IKJ56688I CHAR(0)  LINE(0)   PROMPT  INTERCOM  NOPAUSE MSGID  NOMODE  WTPMS
G  NORECOVER PREFIX(MATEVA)  PLANGUAGE(ENU) SLANGUAGE(ENU) VARSTORAGE(LOW)
IKJ56689I DEFAULT LINE/CHARACTER DELETE CHARACTERS IN EFFECT FOR THIS TERMINAL
READY
█
```

The PROFILE command will display the current settings of my TSO user profile.

THE TSO READY PROMPT

```
File      Options  Keypad

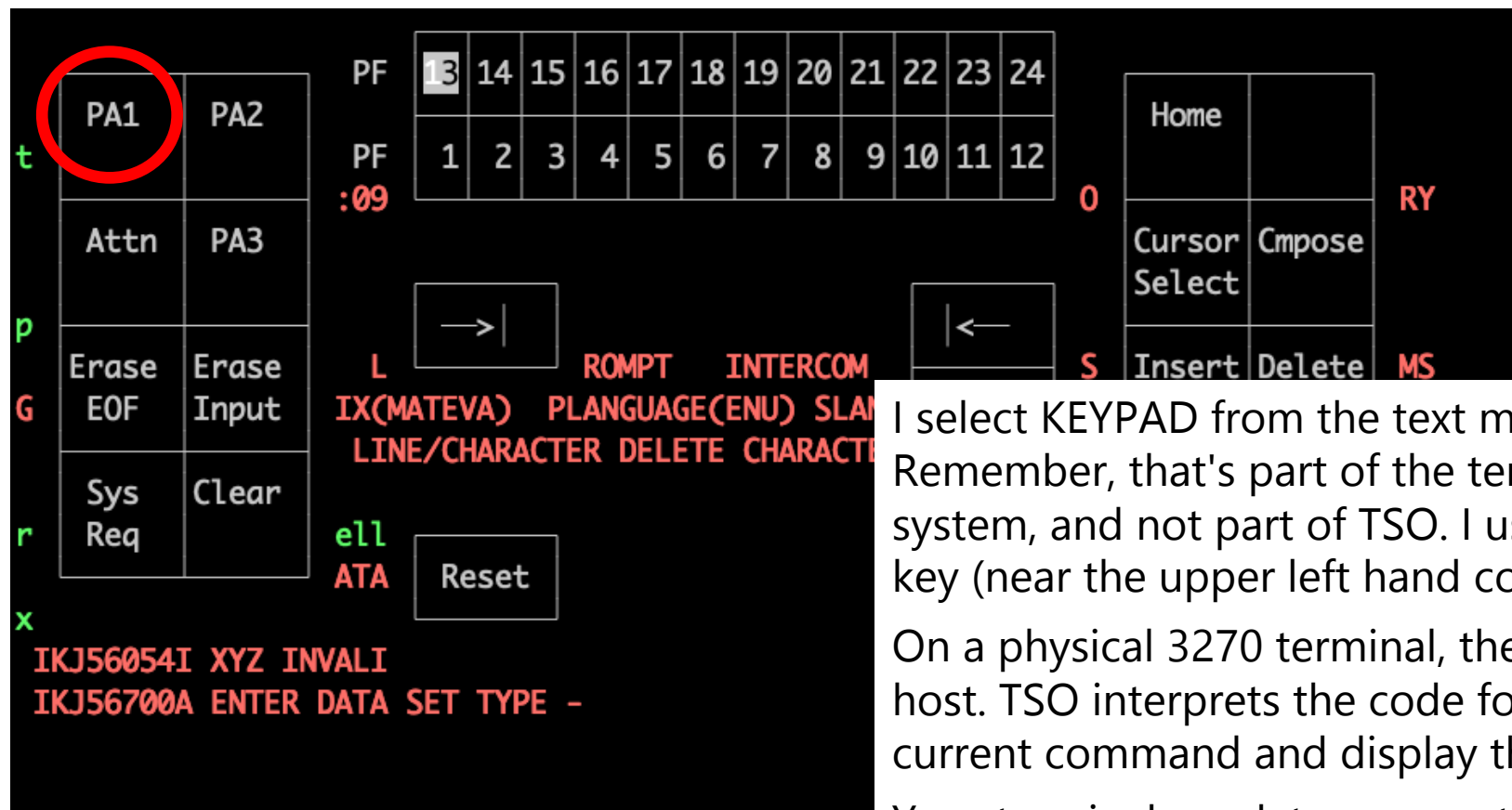
READY
time
IKJ56650I TIME-05:09:12 PM. CPU-00:00:00 SERVICE-1814 SESSION-00:03:23 FEBRUARY
16,2024
READY
profile
IKJ56688I CHAR(0)  LINE(0)   PROMPT  INTERCOM  NOPAUSE MSGID   NOMODE  WTPMS
G  NORECOVER PREFIX(MATEVA)  PLANGUAGE(ENU) SLANGUAGE(ENU) VARSTORAGE(LOW)
IKJ56689I DEFAULT LINE/CHARACTER DELETE CHARACTERS IN EFFECT FOR THIS TERMINAL
READY
run cobol.source(hello)
IKJ56700A ENTER DATA SET TYPE -
xyz
IKJ56054I XYZ INVALID DATA TYPE
IKJ56700A ENTER DATA SET TYPE -
█
```

Now let's make a mistake and see how to interrupt the TSO command in progress.

Here I've entered the RUN command and specified a data set name. TSO prompts for the data set type, and I respond with an invalid value.

I don't remember the valid values, so I want to blow out of this command and try it again later.

THE TSO READY PROMPT



I select KEYPAD from the text menu at the top of the display. Remember, that's part of the terminal emulator on my local system, and not part of TSO. I use the keypad to "press" the PA1 key (near the upper left hand corner).

On a physical 3270 terminal, the PA1 key sends a code to the host. TSO interprets the code for PA1 to mean "interrupt the current command and display the ready prompt".

Your terminal emulator supports simulating a PA1 keypress, even if it doesn't look identical to this example.

In any case, when you get stuck in TSO, PA1 is the way out.

THE TSO READY PROMPT

```
File      Options  Keypad

READY
time
IKJ56650I TIME-05:09:12 PM. CPU-00:00:00 SERVICE-1814 SESSION-00:03:23 FEBRUARY
16,2024
READY
profile
IKJ56688I CHAR(0)  LINE(0)   PROMPT  INTERCOM  NOPAUSE MSGID   NOMODE  WTPMS
G  NORECOVER PREFIX(MATEVA)  PLANGUAGE(ENU) SLANGUAGE(ENU) VARSTORAGE(LOW)
IKJ56689I DEFAULT LINE/CHARACTER DELETE CHARACTERS IN EFFECT FOR THIS TERMINAL
READY
run cobol.source(hello)
IKJ56700A ENTER DATA SET TYPE -
xyz
IKJ56054I XYZ INVALID DATA TYPE
IKJ56700A ENTER DATA SET TYPE -
|
READY
logoff
```

To log off, type LOGOFF and press enter.

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

We don't spend much time using TSO the way we've been demonstrating it. We won't dwell on all the available commands.

Instead, we use a full-screen environment that wraps the TSO commands plus a lot more – ISPF.

ISPF is a huge improvement over basic TSO. It's tremendously customizable and extensible, but still less polished than modern-day IDEs and smart editors.

ISPF is the third of the three pure-text development environments that came from the era before graphical user interfaces were common and have stood the test of time.

You may already use one or both of the others – EMACS and VIM. Those are children of Unix. ISPF is just for z/OS, but in its heyday at least as many developers used ISPF daily as those who used EMACS and VIM.

```

File      Options  Keypad

  Menu  Utilities  Compilers  Options  Status  Help

-----
Option ==> | ISPF Primary Option Menu

0  Settings      Terminal and user parameters      User ID . : MATEVA
1  View          Display source data or listings   Time. . . : 17:31
2  Edit          Create or change source data       Terminal. : 3278
3  Utilities     Perform utility functions         Screen. . : 1
4  Foreground    Interactive language processing   Language. : ENGLISH
5  Batch         Submit job for language processing  Appl ID . : ISR
6  Command       Enter TSO or Workstation commands  TSO logon : MATPROC
7  Dialog Test   Perform dialog testing            TSO prefix: MATEVA
10 SCLM          SW Configuration Library Manager  System ID : MATE
11 Workplace     ISPF Object/Action Workplace      MVS acct. : MT682
12 DITTO         DITTO/ESA for MVS                 Release . : ISPF 6.3
13 FMN          File Manager Operations and Control
15 DB2 V10      DB2 V10 Primary Menu
17 QMF V10      DB2 Query Management Facility
18 Debug Tool   Debug Tool Utility functions
S  SDSF         Spool Search and Display Facility

Enter X to Terminate using log/list defaults

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and F8=Forward F9=Swap
F10=Actions F12=Cancel
  
```

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

```
User ID . : MATEVA
Time. . . : 17:31
Terminal. : 3278
Screen. . : 1
Language. : ENGLISH
App1 ID . : ISR
TSO logon : MATPROC
TSO prefix: MATEVA
System ID : MATE
MVS acct. : MT682
Release . : ISPF 6.3
```

Let's explore this display. We'll start by zooming in on the status information displayed on the right side.

What can we teach ourselves by examining these values?

Here's a small challenge: Find the answers to the following questions in the screenshot.

- 1) What's the system ID of the z/OS instance we're connected to?
- 2) What release of ISPF are we using?
- 3) Which particular 3270 model is our terminal emulator pretending to be?
- 4) Is it possible to have more than one active screen under ISPF?
- 5) Is it possible for ISPF to display information in languages other than English?

(Bonus) Was I working after hours when I took this screenshot?

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

The main part of the display shows the Primary Option Menu. We choose an option from the menu by entering its code in the Option field and pressing Enter.

Before we do that, let's zoom in on the menu and check it out.

Everything on this menu is configurable. However, nearly all z/OS installations leave options 0 through 6 and option S alone. The rest might be different, depending on how the particular z/OS instance is configured and what optional products may be installed on it.

0	Settings	Terminal and user parameters
1	View	Display source data or listings
2	Edit	Create or change source data
3	Utilities	Perform utility functions
4	Foreground	Interactive language processing
5	Batch	Submit job for language processing
6	Command	Enter TSO or Workstation commands
7	Dialog Test	Perform dialog testing
10	SCLM	SW Configuration Library Manager
11	Workplace	ISPF Object/Action Workplace
12	DITTO	DITTO/ESA for MVS
13	FMN	File Manager Operations and Control
15	DB2 V10	DB2 V10 Primary Menu
17	QMF V10	DB2 Query Management Facility
18	Debug Tool	Debug Tool Utility functions
S	SDSF	Spool Search and Display Facility

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

```
File      Options  Keypad

  Menu  Utilities  Compilers  Options  Status  Help

ISPF Primary Option Menu

Option ==> |

0 Settings      Terminal and user parameters      User ID . : MATEVA
1 View          Display source data or listings  Time. . . : 17:51
2 Edit          Create or change source data     Terminal. : 3278
3 Utilities     Perform utility functions         Screen. . : 1
4 Foreground    Interactive language processing    Language. : ENGLISH
5 Batch         Submit job for language processing  Appl ID . : ISR
6 Command       Enter TSO or Workstation commands   TSO logon : MATPROC
7 Dialog Test   Perform dialog testing            TSO prefix: MATEVA
10 SCLM         SW Configuration Library Manager  System ID : MATE
11 Workplace    ISPF Object/Action Workplace      MVS acct. : MT682
12 DITTO        DITTO/ESA for MVS                 Release . : ISPF 6.3
13 FMN          File Manager Operations and Control
15 DB2 V10      DB2 V10 Primary Menu
17 QMF V10      DB2 Query Management Facility
18 Debug Tool   Debug Tool Utility functions
S SDSF          Spool Search and Display Facility

Enter X to Terminate using log/list defaults

F1=Help    F2=Split    F3=Exit    F7=Backward  F8=Forward  F9=Swap
F10=Actions F12=Cancel
```

Along the bottom of the display, there are some hints about what the Program Function (PF) keys do. This is all configurable, but nearly all installations leave these settings alone. These functions are consistent throughout ISPF panels:

PF1 = Help

PF8 = Scroll forward or show the next "thing"

PF7 = Scroll backward or show the previous "thing"

PF3 = Save changes and exit the current panel

PF12 = Cancel changes and exit the current panel

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

Along the top of the display, there's a text menu bar. This one is part of ISPF, not part of the terminal emulator.

The items on this menu correspond to options you can select from the Primary Option Menu or by pressing a PF key. The menu is provided in case you re-map the standard function keys to other functions. You still have a way to reach any option you need.

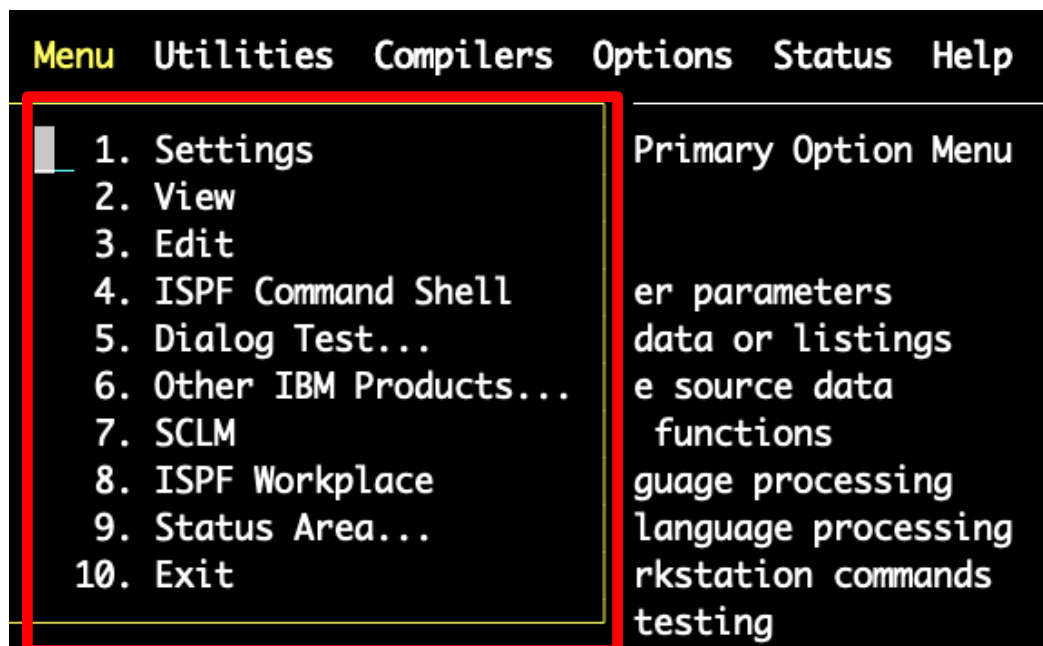
If you're using a terminal emulator on a GUI desktop, you can move the cursor by clicking somewhere in the display window, but ISPF does not respond to pointer gestures. It's a pure text environment. So normally, you'd move the cursor around with the tab key and arrow keys.

```
File      Options  Keypad
Menu Utilities Compilers Options Status Help
ISPFS Primary Option Menu
Option ==> |
0 Settings      Terminal and user parameters      User ID . : MATEVA
1 View          Display source data or listings    Time. . . : 17:51
2 Edit          Create or change source data        Terminal. : 3278
3 Utilities      Perform utility functions          Screen. . : 1
4 Foreground     Interactive language processing     Language. : ENGLISH
5 Batch          Submit job for language processing   Appl ID . : ISR
6 Command        Enter TSO or Workstation commands    TSO logon : MATPROC
7 Dialog Test    Perform dialog testing              TSO prefix: MATEVA
10 SCLM          SW Configuration Library Manager    System ID : MATE
11 Workplace     ISPF Object/Action Workplace        MVS acct. : MT682
12 DITTO         DITTO/ESA for MVS                   Release . : ISPF 6.3
13 FMN           File Manager Operations and Control
15 DB2 V10       DB2 V10 Primary Menu
17 QMF V10       DB2 Query Management Facility
18 Debug Tool    Debug Tool Utility functions
S SDSF          Spool Search and Display Facility

Enter X to Terminate using log/list defaults

F1=Help   F2=Split  F3=Exit   F7=Backward F8=Forward F9=Swap
F10=Actions F12=Cancel
```

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY



Positioning the cursor anywhere on the word, Menu, and pressing Enter causes this pop-up menu to appear. It contains several of the same options as we find on the Primary Option Menu. Those items activate the same functions as the primary options.

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

Positioning the cursor anywhere on the word, Utilities, and pressing Enter causes this pop-up menu to appear. It contains several of the same options we can reach through Primary Menu Option 3, Utilities.

Menu		Utilities	Compilers	Options	Status	Help
Optio		Primary Option Menu				
0	Se	1. Library				
1	Vi	2. Data set				
2	Ed	3. Move/Copy				
3	Ut	4. Data Set List				
4	Fo	5. Reset Statistics				
5	Ba	6. Hardcopy				
6	Co	7. Download...				
7	Di	8. Outlist				
10	SC	9. Commands...				
11	Wo	*0. Reserved				
12	DI	11. Format				
13	FM	12. SuperC				
15	DB	13. SuperCE				
		14. Search-For				
		15. Search-ForE				

LAB 4: EXPLORING TSO AND ISPF

Follow the instructions for Lab 4 in `labs/labs.md` in the course repo.

HOW MUCH DO YOU NEED TO KNOW ABOUT ISPF?

As an infrastructure engineer who performs most of the work using off-platform tools like Ansible, you don't have to become an expert with TSO and ISPF.

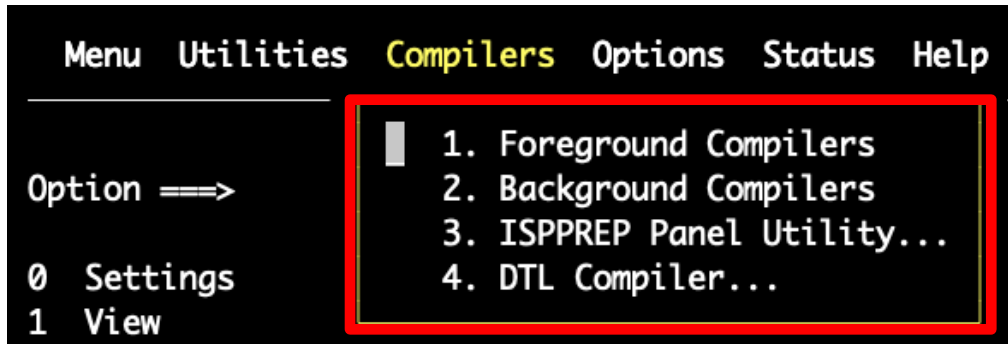
Because Ansible is a leaky abstraction over z/OS system commands, you will have to sign on directly to a z/OS instance occasionally to verify the effects of your Ansible playbooks or to troubleshoot issues.

For that, you need a basic understanding of TSO and ISPF sufficient to

- find data sets and view their attributes;
- create and delete data sets;
- perform basic editing operations on data sets;
- submit jobs;
- view and delete the output from jobs (SYSOUT, SYSPRINT); and
- position the cursor to enable your 3270 emulator to transfer files.

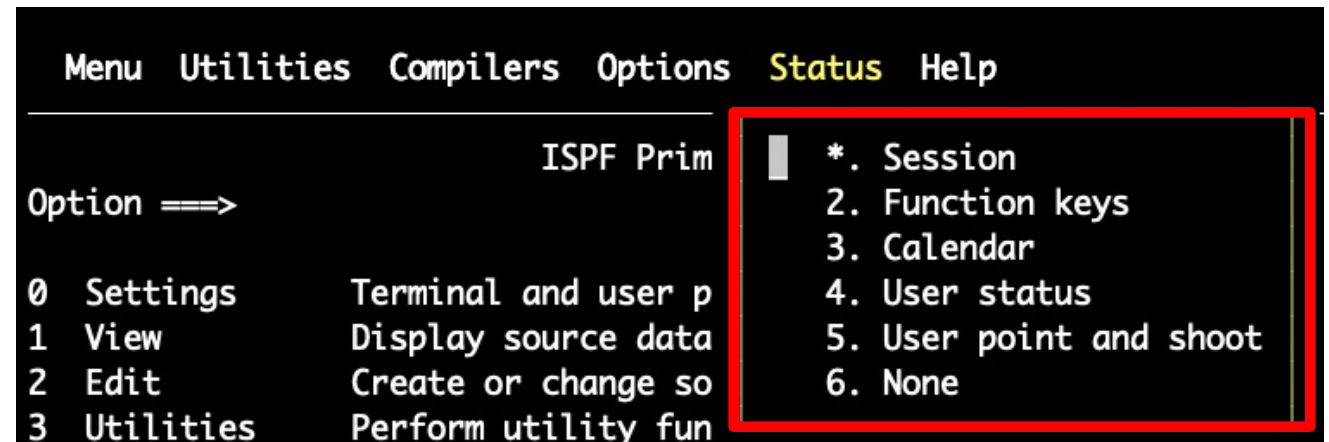
We will practice those tasks in the lab environment. The remainder of the slides in this deck are for future reference if you need them.

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY



The Compilers menu item offers some of the same functions as the Foreground Primary Option.

The Status menu item contains assorted options that aren't necessarily related to each other. Here you can customize function keys and view or set a few options.



ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

The Status/Calendar menu item toggles a calendar widget over the area on the main ISPF panel where session status usually appears.

```
Menu Utilities Compilers Options Status Help
ISPF Primary Option Menu
Option ==> █

0 Settings      Terminal and user parameters
1 Background    Interactive language processing
5 Batch         Submit job for language processing
6 Command       Enter TSO or Workstation commands
7 Dialog Test   Perform dialog testing
10 SCLM         SW Configuration Library Manager
11 Workplace    ISPF Object/Action Workplace
12 DITTO        DITTO/ESA for MVS
13 FMN         File Manager Operations and Control
15 DB2 V10      DB2 V10 Primary Menu
17 QMF V10      DB2 Query Management Facility
18 Debug Tool   Debug Tool Utility functions
S SDSF         Spool Search and Display Facility

Enter X to Terminate using log/list defaults
```

< Calendar >
February 2024
Su Mo Tu We Th Fr Sa
1 2 3
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29
Time : 07:42
Day of year. : 047

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

The Help menu item gives you the same options as you get by pressing PF1 (based on default function key mapping).

The screenshot displays the ISPF Main Menu at the top with options: Menu, Utilities, Compilers, Options, Status, and Help (highlighted in yellow). Below this is a table of menu items. To the right, the Help menu is shown, listing 20 options from 1. General to 20. Index, which is enclosed in a red rectangular border.

	Menu	Utilities	Compilers	Options	Status	Help
0	Settings	Terminal and user parameter				
1	View	Display source data or list				
2	Edit	Create or change source dat				
3	Utilities	Perform utility functions				
4	Foreground	Interactive language proces				
5	Batch	Submit job for language pro				
6	Command	Enter TSO or Workstation co				
7	Dialog Test	Perform dialog testing				
10	SCLM	SW Configuration Library Ma				
11	Workplace	ISPF Object/Action Workplac				
12	DITTO	DITTO/ESA for MVS				
13	FMN	File Manager Operations and				
15	DB2 V10	DB2 V10 Primary Menu				
17	QMF V10	DB2 Query Management Facili				
18	Debug Tool	Debug Tool Utility function				
S	SDSF	Spool Search and Display Fa				

Enter X to Terminate using log/list def

1. General
2. Settings
3. View
4. Edit
5. Utilities
6. Foreground
7. Batch
8. Command
9. Dialog Test
10. LM Facility
11. IBM Products
12. SCLM
13. Workplace
14. Exit
15. Status Area
16. About...
17. Changes for this Release
18. Tutorial
19. Appendices
20. Index

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

```
Log/List  Function keys  Colors  Environ  Workstation  Identifier  Help

ISPFI Settings

Command ==> |

Options                                         More:  +
Enter "/" to select option
- Command line at bottom
/ Panel display CUA mode
/ Long message in pop-up
- Tab to action bar choices
- Tab to point-and-shoot fields
/ Restore TEST/TRACE options
- Session Manager mode
/ Jump from leader dots
- Edit PRINTDS Command
/ Always show split line
- Enable EURO sign

Print Graphics
Family printer type 2
Device name . . . .
Aspect ratio . . . 0

General
Input field pad . . B
Command delimiter . ;

Member list options
Enter "/" to select option
/ Scroll member list
- Allow empty member list
- Allow empty member list (nomatch)
/ Empty member list for edit only

Terminal Characteristics
Screen format  2  1. Data    2. Std    3. Max    4. Part

Terminal Type  3  1. 3277    2. 3277A   3. 3278    4. 3278A
F1=Help      F2=Split  F3=Exit   F7=Backward F8=Forward F9=Swap
F10=Actions  F12=Cancel
```

Option 0, Settings, from the Primary Options Menu, takes you to the ISPF Settings panel where you can adjust some of the interactive behavior of ISPF.

You don't need to memorize any of this just now, but be aware this option is where you can customize your ISPF user experience a little.

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

ISPF is where we spend most of our time in the course of a typical software developer's day.

We use most of the standard options on the Primary Option Menu, but we use the **Edit**, **Utilities**, and **SDSF** options most heavily.

You will become familiar with these bit by bit in the context of demonstrations and lab exercises.

```

Menu Utilities Compilers Options Status Help
ISPF Primary Option Menu
Option ==> █
0 Settings Terminal and user parameters
1 View Display source data or listings
2 Edit Create or change source data
3 Utilities Perform utility functions
4 Foreground Interactive language processing
5 Batch Submit job for language processing
6 Command Enter TSO or Workstation commands
7 Dialog Test Perform dialog testing
10 SCLM SW Configuration Library Manager
11 Workplace ISPF Object/Action Workplace
12 DITTO DITTO/ESA for MVS
13 FMN File Manager Operations and Control
15 DB2 V10 DB2 V10 Primary Menu
17 QMF V10 DB2 Query Management Facility
18 Debug Tool Debug Tool Utility functions
S SDSF Spool Search and Display Facility
Enter X to Terminate using log/list defaults

```

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

Those who work with DB2 also use functionality under the **DB2** and **QMF** (Query Management Facility) options.

```

Menu Utilities Compilers Options Status Help
ISPF Primary Option Menu
Option ==> █
0 Settings Terminal and user parameters
1 View Display source data or listings
2 Edit Create or change source data
3 Utilities Perform utility functions
4 Foreground Interactive language processing
5 Batch Submit job for language processing
6 Command Enter TSO or Workstation commands
7 Dialog Test Perform dialog testing
10 SCLM SW Configuration Library Manager
11 Workplace ISPF Object/Action Workplace
12 DITTO DITTO/ESA for MVS
13 FMN File Manager Operations and Control
15 DB2 V10 DB2 V10 Primary Menu
17 QMF V10 DB2 Query Management Facility
18 Debug Tool Debug Tool Utility functions
S SDSF Spool Search and Display Facility

Enter X to Terminate using log/list defaults

```


WHAT YOU KNOW AND WHAT YOU CAN DO

- You can log on and off TSO.
- You can start and exit ISPF.
- You can navigate around ISPF using the menu bar ("action bar") and the Primary Option Menu.
- You know which ISPF option to select if you want to customize the behavior of ISPF.

WHAT'S NEXT

- Creating a z/OS Data Set